

Exercițiul 1.

(3 puncte)

Pentru $z \in \mathbb{C}$, notăm $z = x + iy$, $x, y \in \mathbb{R}$ și avem identificarea $z \simeq (x, y)$. Considerăm funcția $u : \mathbb{C} \rightarrow \mathbb{R}$, $u(x, y) = x^2 - y^2 + 2e^{-y} \cos x$. Găsiți o funcție olomorfă $f : \mathbb{C} \rightarrow \mathbb{C}$ pentru care $\operatorname{Re}(f) = u$.

Exercițiul 2.

(3 puncte)

Folosind teorema reziduurilor, calculați următoarea integrală:

$$\int_{-\infty}^{\infty} \frac{3x^2 + 1}{1 + x^4} dx.$$

Exercițiul 3.

(3 puncte)

Considerăm semi-banda orizontală $L = \{z : z = x + iy, x > 0, y \in (-\pi, 0)\}$. Găsiți un șir de aplicații conforme bijective care transformă L în H^+ .

Exercițiul 4.

(3 puncte)

Fie $f : \mathbb{C} \rightarrow \mathbb{C}$ olomorfă. Demonstrați că dacă $g(z) = f(\bar{z})$ este tot olomorfă, atunci f este constantă.

① $f = ?$ $\operatorname{Re}(f) = u$ $u(x, y) = x^2 - y^2 + 2e^{-y} \cos x$.

Soluție:

Arată că u este armonică.

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} (u) = \frac{\partial}{\partial x} (2x - 2e^{-y} \sin x) +$$

$$+ \frac{\partial}{\partial y} (-2y - 2e^{-y} \cos x) = 2 - 2e^{-y} \cos x + (-2 + 2e^{-y} \cos x) = 0.$$

Deci u este armonică. $\exists$ $f \in H(\mathbb{C})$ a.i. $\operatorname{Re} f = u$.

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \stackrel{\text{C-R}}{=} \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y} =$$

$$= 2x - 2e^{-y} \sin x - i(-2y - 2e^{-y} \cos x)$$

Fie $y=0 \Rightarrow z = x + iy = x \Rightarrow f'(x) = 2x - 2 \sin x - i(-2 \cos x) \Rightarrow f'(x) = 2x - 2 \sin x + i 2 \cos x.$

$$\int (2x - 2 \sin x + i 2 \cos x) dx = x^2 + 2 \cos x + 2i \sin x + C.$$

Fie $g(z): \mathbb{C} \rightarrow \mathbb{C}; g(z) = z^2 + 2 \cos z + 2i \sin z$

Avem că $g'(x) = f'(x)$

$R \subseteq \mathbb{C}$ are puncte de acumulare
 $g \in H(\mathbb{C}) \Rightarrow$ din Teorema de identitate a funcțiilor holomorfe

$$f' = g' \text{ pe } \mathbb{C} \Rightarrow f(z) = g(z) + C \text{ i.e.}$$

$$f(z) = z^2 + 2 \cos z + 2i \sin z + C.$$

②. $\int_{-\infty}^{\infty} \frac{3x^2+1}{1+x^4} dx$

~~2-2-~~
aream rădăcinile
 $1+x^4=0$

$$z_0 = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4}$$

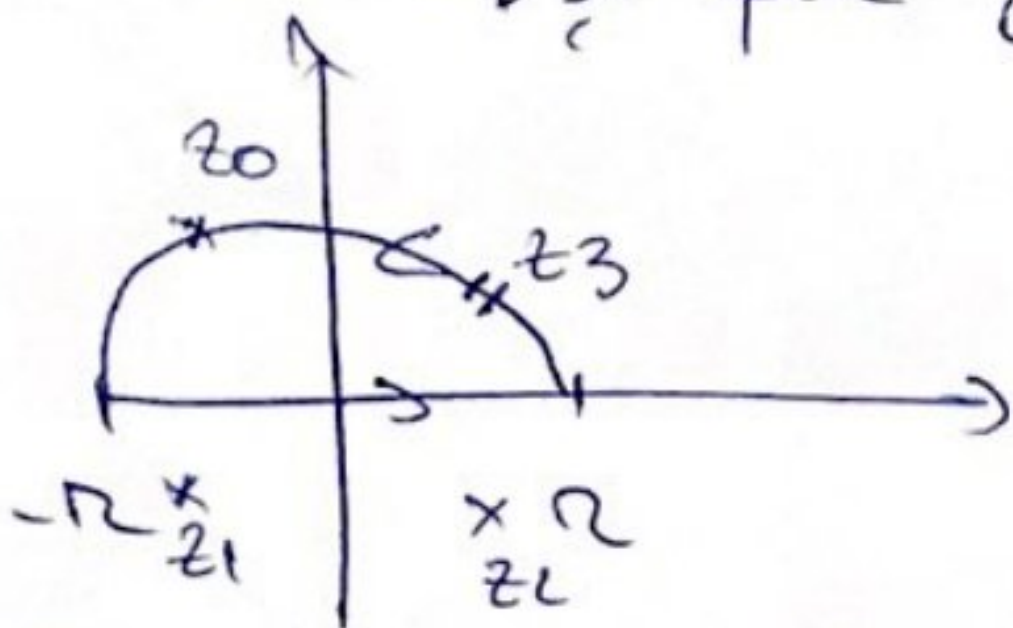
$$z_1 = \cos \left(\frac{3\pi}{4} \right) + i \sin \left(\frac{3\pi}{4} \right)$$

$$z_2 = \cos \left(\frac{5\pi}{4} \right) + i \sin \left(\frac{5\pi}{4} \right)$$

$$z_3 = \cos \left(\frac{7\pi}{4} \right) + i \sin \left(\frac{7\pi}{4} \right)$$

Fie $f: \mathbb{C} \setminus \{z_0, z_1, z_2, z_3\} \rightarrow \mathbb{C}$ $f(z) = \frac{3z^2+1}{1+z^4}$

Fie $n > 0$, și fie $\gamma_n: [0, 1] \rightarrow \mathbb{C}$; $\gamma_n(t) = e^{i\pi t}$.



$$\Gamma_n = [-R, R] \cup \gamma_n(t).$$

$f \in H(\mathbb{C} \setminus D)$. D fam. finită de puncte singulare izolate. și $(\mathbb{C} \setminus D) \cup D = \mathbb{C} \quad \forall n > 0, \quad \Gamma_n \sim 0$

~~$$i(\Gamma_n, z_0) = 1; i(\Gamma_n, z_1) = 0; i(\Gamma_n, z_2) = 0; i(\Gamma_n, z_3) = 0$$~~

$$i(\Gamma_n, z_0) = 1; i(\Gamma_n, z_1) = 0; i(\Gamma_n, z_2) = 0; i(\Gamma_n, z_3) = 0$$

$$f(z) = \frac{3z^2+1}{(z-z_1)(z-z_2)(z-z_3)} \rightarrow g(z)$$

$g(z_0) \neq 0 \Rightarrow z_0$ pol de ordin 1. ~~te~~

Analog, arătăm pt z_3 că este pol de ordin 1. ($g(z_0) \neq 0, 3z_0^2+1 \neq 0$.)

$$\int_{\Gamma_n} f(z) dz = \int_{\gamma_n} f(z) dz + \int_{-R}^R f(x) dx$$

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Deci, Avem:

$$\int_{-\infty}^{\infty} \frac{3x^2+1}{1+x^4} dx = (\text{Res}(f, z_0) + \text{Res}(f, z_3)) 2\pi i \quad (1)$$

z_0 pol de ordin 1 $\Rightarrow$

$$\text{Res}(f, z_0) = \lim_{z \rightarrow z_0} \frac{1}{0!} \frac{d^{1-1}}{dz^{1-1}} \left((z-z_0)^1 \cdot f(z) \right) =$$

$$= \lim_{z \rightarrow z_0} 1 \cdot (z-z_0) \cdot \frac{3z^2+1}{(z-z_0)(z-z_1)(z-z_2)(z-z_3)}$$

$$= \lim_{z \rightarrow z_0} \frac{3z^2+1}{(z-z_1)(z-z_2)(z-z_3)} = \frac{3z_0^2+1}{(z_0^2-z_1^2)(z_0-z_2)(z_0-z_3)}$$

$$= \frac{6(1+i)^2 + 4}{4\sqrt{2}(1+i) \cdot \frac{4i^2}{2}} = \frac{3(1+i)^2 + 2}{2\sqrt{2}(1+i) \cdot i}$$

$$\text{Res}(f, z_3) = \lim_{z \rightarrow z_3} (z-z_3) \cdot \frac{3z^2+1}{(z-z_0)(z-z_1)(z-z_2)(z-z_3)} =$$

$$\lim_{z \rightarrow z_3} \frac{3z^2+1}{(z-z_0)(z-z_1)(z-z_2)} = \frac{3z_3^2+1}{(z_3-z_0)(z_3-z_1)(z_3-z_2)} =$$

$$= \frac{3z_3^2+1}{(z_3-z_0)(z_3-z_1)(z_3+z_1)} = \frac{3z_3^2+1}{(z_3-z_0)(z_3+z_0)(-2z_3)} =$$

$$= \frac{3z_3^2+1}{(z_3^2-z_0^2)(-2z_3)} = \frac{3 \cdot \frac{1}{4}(-1+i)^2 + 1}{(z_3^2-z_0^2)(-2z_3)} =$$

$$= \frac{3 \cdot \frac{1}{4}(-1+i)^2 + 1}{\left(\frac{(-1+i)^2 - (1+i)^2}{4} \right) \sqrt{2}(-1+i)} =$$

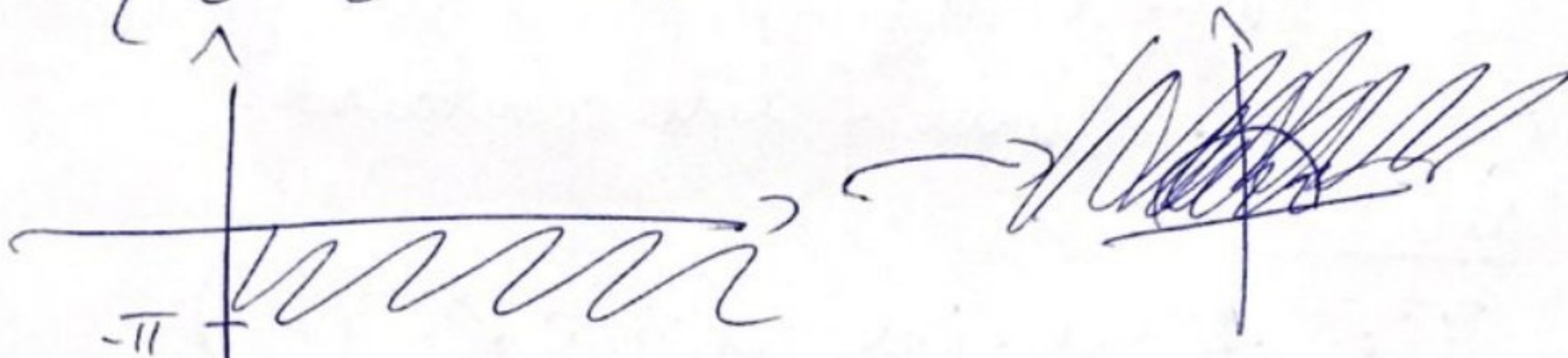
$$= \frac{3(-1+i)^2 + 2}{-4i\sqrt{2}(-1+i)}$$

Ans - 4 -

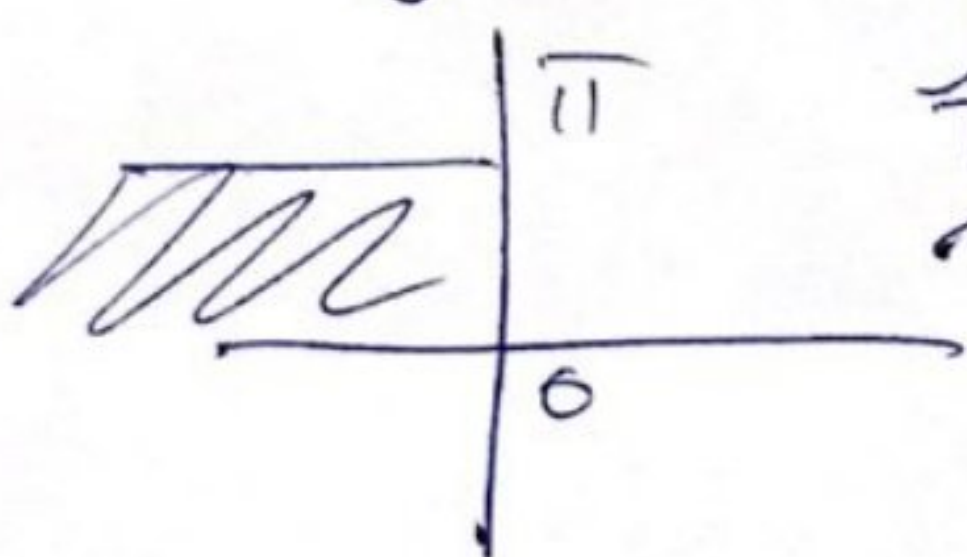
Soln

$$\begin{aligned}
 \int_{-\infty}^{\infty} \frac{3x^2+1}{1+x^4} dx &= 2\pi i \left(\frac{3(1+i)^2+2}{2\sqrt{2}(1+i) \cdot i} + \frac{3(-1+i)^2+2}{-4i(-1+i)\sqrt{2}} \right) \\
 &= \left(\frac{3 \cdot 2i + 2}{2\sqrt{2}(1+i) \cdot i} + \frac{3(-2i) + 2}{-4i(-1+i)\sqrt{2}} \right) 2\pi i = \\
 &= \left(\frac{6i+2}{\sqrt{2}(1+i)} + \frac{-6i+2}{(-1+i)\sqrt{2}} \right) \pi = \\
 &= \left(\frac{(6i+2)(-1+i) + (-6i+2)(1+i)}{(1+i)(-1+i)} \right) \frac{\pi}{\sqrt{2}} = \\
 &= \left(\frac{-6i - 6 - 2 + 2i - 6i + 6 + 2 + 2i}{(1+i)(-1+i)} \right) \frac{\pi}{\sqrt{2}} = \\
 &= \left(\frac{-12i + 4i}{-2} \right) \frac{\pi}{\sqrt{2}} = \frac{4i\pi}{\sqrt{2}}
 \end{aligned}$$

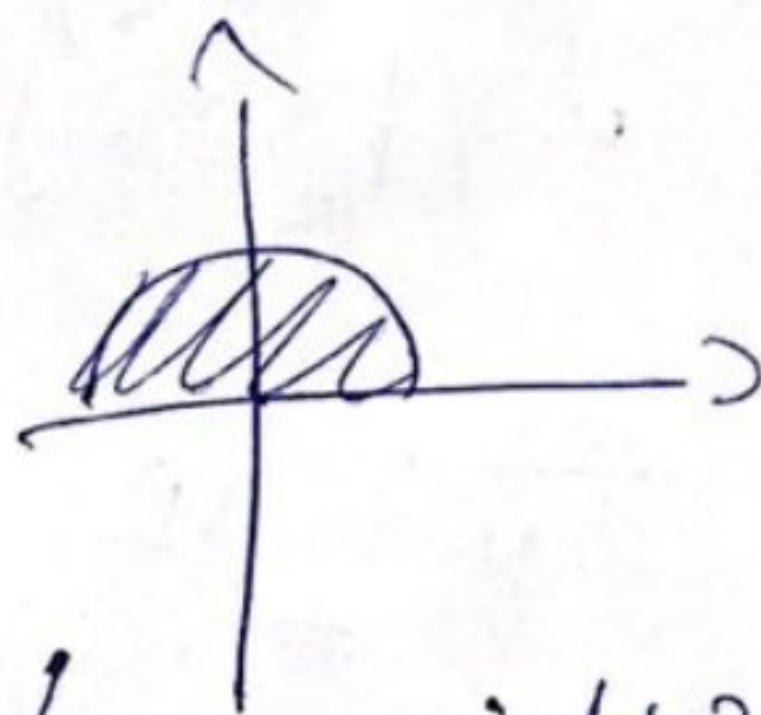
③ $L = \{z: z = x + iy, x > 0, y \in (-\pi, 0)\}$



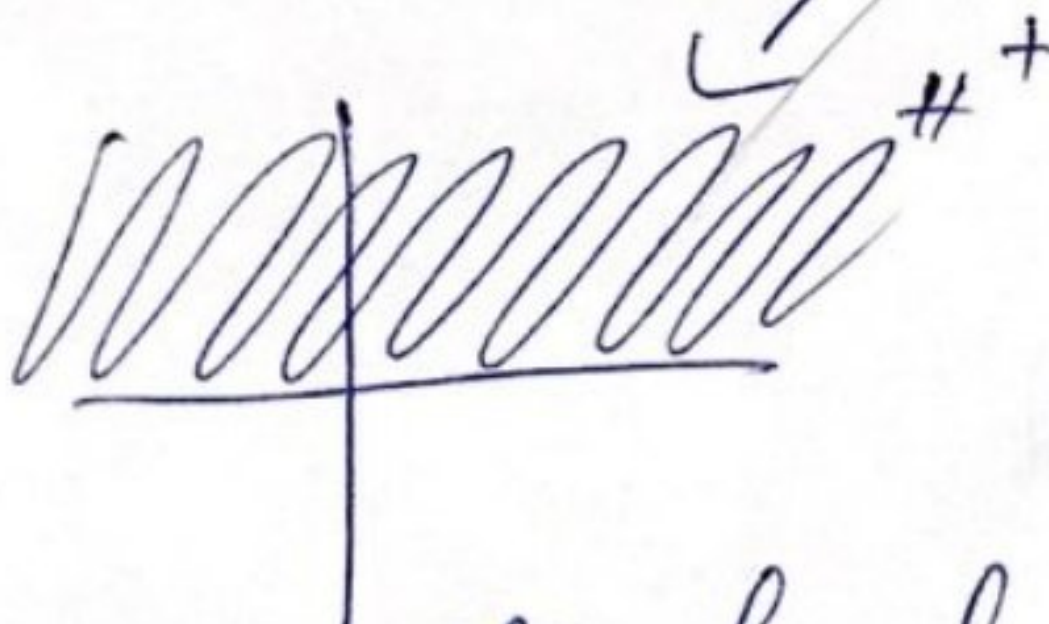
$f_1(z) = -z$



$f_2(z) = e^z$



$f_3(z) = i \frac{1+z}{1-z}$



Deci sirul este f_1, f_2, f_3 .

④

Fie $f: \mathbb{C} \rightarrow \mathbb{C}$ o morfă. Dacă $g(z) = f(\bar{z})$ o morfă, atunci f este constantă.

Soluție:

Fie $f = u + iv$, $u, v \in C^2(\mathbb{R})$.

$z = x + iy$ $f(z) = u + iv \Rightarrow$ ~~$f(z)$~~

Stim că $f \in H(\mathbb{C}) \Rightarrow f(\bar{z}) = \overline{f(z)}$.

$$\begin{cases} \frac{\partial u}{\partial x} = + \frac{\partial v}{\partial y} & (1) \\ \frac{\partial u}{\partial y} = - \frac{\partial v}{\partial x} & (2) \end{cases}$$

~~$f = u + iv$~~ $u - iv$.

$$\begin{cases} \frac{\partial u}{\partial x} = - \frac{\partial v}{\partial y} & (3) \\ \frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} & (4) \end{cases}$$

Din (1) și (3) $\Rightarrow \frac{\partial v}{\partial y} = - \frac{\partial v}{\partial y} \Rightarrow 2 \frac{\partial v}{\partial y} = 0 \Rightarrow \frac{\partial v}{\partial y} = 0$ Deci v este constantă.

Din (2) și (4) $\Rightarrow \frac{\partial u}{\partial y} = - \frac{\partial u}{\partial y} \Rightarrow 2 \frac{\partial u}{\partial y} = 0 \Rightarrow \frac{\partial u}{\partial y} = 0 \Rightarrow u$ este constantă

Deci și f este constantă complexă.

Cazul când $f|_{\mathbb{R}} \notin \mathbb{R}$.

Fig, $f \in H(\mathbb{D}) \Rightarrow f$ admite dezvoltare
în serie de puteri în jurul lui z_0 .

$$f(z) = \sum a_n (z - z_0)^n$$

$$f(\bar{z}) = \sum a_n (\bar{z} - z_0)^n.$$

$$f(z) - f(\bar{z}) = \sum a_n ((z - z_0)^n - (\bar{z} - z_0)^n).$$

$$a_n \neq 0 \Rightarrow (z - z_0)^n - (\bar{z} - z_0)^n = 0 \Rightarrow$$

$$(z - z_0)^n = \bar{z} -$$

$$\frac{f(z) - f(\bar{z})}{a_n} = (z - z_0)^n - (\bar{z} - z_0)^n$$

$$\overline{f(\bar{z})} = 2a_n (z - \bar{z}_0)^n. \quad ?$$